MIL-T-13366 (ORD)
5 April 1954

MILITARY SPECIFICATION

TUNGSTEN CARBIDE, CRYSTALLINE

1. SCOPE

1.1 This specification covers crystalline tungsten carbide for use in the manufacture of projectile cores. The tungsten carbide is produced directly from low-grade tungsten bearing materials and is not suitable for production applications without reduction in particle size.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-C-2550 - General Specification For Ammunition Except Small Arms Ammunition
MIL-C-6054 - Containers: Metal, Reusable, Exterior Shipping

STANDARDS

MILITARY

MIL-STD-129 - Marking of Shipments

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Crystalline tungsten carbide shall comply with the chemical composition specified in Table I, when determined as specified in 4.4.

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TABLE I

Ingredient	Percent
Total carbon	6.10 ± 0.10
Free carbon, maximum	0.10
Cobalt, maximum	0.50
Iron, maximum	0.25
Molybdenum, maximum	0.25
Tantalum and Columbium, maximum	1.00
Titanium, maximum	0.15
Silicon, maximum	0.05
Oxygen, maximum	0.10
Tungsten, minimum	92.00

3.2 Moisture content.- The moisture content of the material shall not exceed .05 percent when determined as specified in 4.5.

3.3 Granulation.- No particle size is specified due to the necessity of reduction in size as required, prior to use.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.- A lot shall consist of material subjected to the same processing operation and conditions and manufactured in one continuous process.

4.2 Sampling.- Ten percent of the containers in the lot, but in no case more than 10 containers nor fewer than 3, shall be selected by the inspector so as to be representative of the lot. If there are fewer than 3 containers in the lot, all the containers shall be sampled. A representative sample of approximately 1 pound shall be taken from each selected container. Each of these primary samples shall be placed in an airtight container and labeled so that the container from which it was taken can be identified. A composite sample of approximately 1 pound shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight container and labeled to show the name of the material, manufacturer, plant, contract or purchase order number, lot number and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform the inspector may require that any primary sample be tested for compliance with the requirement of this specification. All primary samples shall be held for possible future examination should the composite sample fail to comply with the requirements.

4.3 Inspection.- Inspection shall be as specified in Specification MIL-G-2550.

4.4 Chemical analysis.- Chemical analysis for crystalline tungsten carbide shall be made by standard wet chemical methods (see 6.2) or spectrographic methods (see 6.3) to determine compliance with 3.1. In case of dispute

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between the contractors test results and Government acceptance inspection, chemical analysis may be reconducted using the standard wet chemical method, upon agreement to that effect by the contracting agency and the contractor.

4.5 Determination of moisture content.- An accurately weighed portion of approximately 20 gms of the material shall be transferred to a previously dried tared weighing dish fitted with a cover. The cover shall be removed and the dish, contents, and cover shall be placed in an oven at a controlled temperature of 105° to 110°C for 2 hours. The dish, contents, and cover shall be removed from the oven, and the cover shall be replaced on the dish. The covered dish and contents shall be cooled in a desiccator and weighed. The loss in weight shall be calculated to percent moisture to determine compliance with 3.2.

4.6 Resubmission and retest.- If the composite sample or any primary sample subjected to test, fails to pass the tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from all or any of the containers in the lot at no expense to the Government. The contractor may then remove the defective portions of the lot and resubmit the lot for acceptance. The resubmitted lot shall be accepted, provided that the new samples, selected in accordance with 4.2, pass all the tests required by this specification and that three additional primary samples pass any test or tests failed by the material on original submission.

5. PREPARATION FOR DELIVERY

5.1 Packing.- The material shall be packed in 15 gallon steel containers complying with Specification MIL-C-6054.

5.2 Marking.- In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification

6.2 Information relative to "Wet Chemical Methods" may be obtained from the American Chemical Society's "Analytical Chemistry", 1948, Volume 20, pages 563-570.

6.3 Information relative to "Spectrographic Analysis" may be obtained from an article by John C. Redmond in the periodical "Steel", 1948, Volume 122, Number 20, pages 86-88.

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Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:

Army-Ordnance Corps

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEM PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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